

Technical Note

Viability study of grid-connected solar PV system in Ethiopia



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ABSTRACT

This study explored the potential of grid-connected solar PV power generation in Ethiopia. Overall, 35 locations were assessed for their technical potential considering a 5 MW PV power plant in each site. Input data sources for the study include the National Meteorological Agency of Ethiopia and the Surface Meteorology and Solar Energy Dataset of NASA. Both HOMER and RETScreen software were employed in the study. It was found in the study that the average value of PV power plant capacity factor of the different locations considered is 19.8%, and the mean value for the electricity exported to the grid is 8674 MWh/year. Furthermore, economic viability study of a 5 MW PV grid-connected power plant in Addis Ababa area was further conducted. Financial indicators showed that the proposed solar PV system is economically viable, but it may not be sufficiently attractive for commercial investors unless an incentive mechanism is introduced. Further scenario analysis indicated that the upcoming feed-in tariff law for solar PV power generation (which is not yet introduced) may influence the investment decision of the private sector. With the proposed system, up to 7658 MWh electricity can be generated and a minimum of 1089 tons of greenhouse gas emission can be reduced annually.

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Introduction

According to the 2009/2010 energy balance report, consumption of energy in Ethiopia was about 1.3EJ in which biomass fuels take the highest share (92%), followed by hydrocarbons (7%), and electricity (1%). Out of the total electricity generated in 2009/2010, hydropower accounts for the majority, 88.5%, thermal, 10.9%, and geothermal, 0.6% [1]. So far, the critical mass of Ethiopian population has remained un-electrified. While still having poor energy access at home, Ethiopia aspires to be a “renewable energy hub in East Africa” [2]. Ethiopia heavily relies on its hydropower potential and expands its investment in this sector, including the on-going construction of the largest hydroelectric power dam on Blue Nile River (maximum planned installed capacity of 6000 MW). Though it is well endowed with a wide array of renewable energy resources, Ethiopia has shown little progress in exploiting the opportunity. Among others, its solar energy potential has got tiny attention, and patchy efforts of exploitation of this potential were registered [3]. Adoption of solar photovoltaics (PV) technologies remained so sluggish, and tangible support from the government has been required. Critical adoption barriers, such as lack of awareness and lack of finance, prevail both at the demand and the supply side. Relatively, there are limited but promising

efforts in off-grid rural electrification through PV systems [3]. However, there is plenty of opportunity in supporting the energy access of the nation through both on-grid and off-grid solar PV systems. Often grid-connected systems are connected and feed power into the public electric grid; such systems can be either distributed type, serving a certain grid-connected customer or centralized type, serving as centralized power station and feeding into a transmission grid [4]. Both types of these systems are potential options for countries like Ethiopia.

While the main focus of this study is to analyze the viability of a 5 MW grid-connected solar PV power plant in Addis Ababa area, other potential locations are considered as part of showing the technical potential of PV grid-tied plants throughout the country. So far, such locations have been kept away from feasibility study plans and as a result, less information is available to the private sector. In the next section, brief overview of previous studies and historical background of PV systems in Ethiopia is included.

Previous studies on PV systems in Ethiopia

The first standalone solar PV system in Ethiopia was introduced in the mid of 1980s to a remote village located in the central part of the country [5]. It was a 10.5 kWp PV system installed in the village as a mini-grid system to the villagers, and it was by then claimed to be “the largest of its kind in sub-Saharan Africa” [5, p. 728]. The PV system was installed in an area of 3000 m² serving 390 households.

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Nomenclature

B–C	benefit–cost	IRR	internal rate of return
BOS	balance of system	LCOE	levelized cost of electricity
EEPCo	Ethiopian Electric Power Corporation	NPV	net present value
FIT	feed-in tariff	PV	photovoltaics
GHG	Greenhouse gas	PVGIS	Photovoltaic Geographical Information System
GoE	Government of Ethiopia	SWERA	Solar and Wind Energy Resources Assessment

Each household was provided with a 20 W lamp to get a lighting service from the centralized power for about 4 h in the night. In addition, the village had a street light, a school and a clinic utilizing the solar generated power for lighting and vaccine refrigeration [5]. After serving for few years, the standalone PV system was out of function for lack of maintenance and follow up service [6]. Since then, there has been no comparable mini-grid PV plant in the country.

Further studies in the 1990s were conducted and showed that solar energy is an abundant and potential resource throughout the country [7]. Few pilot studies on the performance of small scale PV systems were carried out, and it was recommended that PV systems were cost-competitive alternatives for rural electrification [8]. But uptake since then has remained so slow. The Ethiopian Electricity Proclamation (No. 86/97) issued in 1997 had stated that any person or corporate entity could apply for a license for electricity production and distribution, with a restriction that a private investment in power plants should not exceed 25 MW [9,10]. However, there was no private investment in solar power plants in Ethiopia. Mainly the Ethiopian Electric Power Corporation (EEPCo) has been a state-owned and vertically integrated monopoly that controls the market from generation to selling of electricity throughout the country [10].

After quite a period of ‘silence’ in solar energy technology adoption in Ethiopia, the Government of Ethiopia established an autonomous institution called Ethiopian Rural Energy Development and Promotion Centre in 2002 with the objective of making a conducive environment for the development and promotion of rural energy technologies including solar energy technologies. However, there were many challenges and barriers to the adoption of solar PV systems which included: high tariff on PV panels, high initial cost of renewable energy systems, lack of knowledge about renewable energy systems, and lack of locally produced system components [9]. All of these challenges are still prevalent in the supply chain of PV system in the country [3].

Later in 2004, the global Solar and Wind Energy Resources Assessment (SWERA) team assessed and released the solar and wind resource potential of Ethiopia. The team produced a solar map of the country (as shown in Fig. 1). A recent solar map of Ethiopia produced by Photovoltaic Geographical Information System (PVGIS) project team is also available online [11]. The PVGIS map represents yearly sum of global irradiation on an optimally inclined surface based on the average data of the period 1998–2013.

Bekele and Palm [12] studied the solar energy potential of four locations in Ethiopia, including Addis Ababa, the capital city. Bekele and Boneya [13] further showed how a PV-wind hybrid system is feasible to electrify a rural village. Most of the other PV related studies focused on off-grid application of solar PV systems [3]. Deichmann et al. [14] also conducted a study on the economics of grid connected supply versus off-grid supply in selected African countries including Ethiopia. In the Ethiopian case, they found that the cost of solar PV generated electricity showed large variability across different areas ranging from about 66 cents/kWh to more than one dollar [14, p. 222].

In general, very limited studies on the cost of solar electricity in sub-Saharan African countries (including Ethiopia) have been conducted [15]. Such previous studies in East Africa are briefly discussed in [15]. Overall, there is an urge for conducting contextual studies so as to inform policy makers and the private sector in each country and in the region at large. Ondraczek [15] further argued that the prevalent understanding that “solar energy costs too high” in this region needs to be challenged through further studies.

As far as the author knowledge is concerned, only a recent state-sponsored pre-feasibility study on solar energy potential of Ethiopia suggested four sites for solar PV grid-connected power plants [16]. In this study, we then tried to assess the potential of 35 locations for grid-tied PV systems in Ethiopia and conducted a viability study of a 5 MW PV grid-connected power plant in Addis Ababa area.

Solar energy potential of Ethiopia: Selected locations

A recent study indicated that Ethiopia, often claiming itself as a nation with “13 months of sunshine¹”, has a potential of an average annual solar radiation energy density of unit area amounting 1992.2 kWh/(m²a) and annual total solar energy reserve of 2,199,000 TW h/a [16]. As part of showing the grid-connected PV power potential, 35 different locations throughout Ethiopia are considered in this study with a typical 5 MW solar PV system in each site.

RETScreen was used to analyze and compare the potential of these sites. RETScreen software is a decision support tool used to analyze and evaluate energy production and savings, emission reductions, life cycle costs and financial feasibility for various types of renewable energy technologies. The software uses the monthly average values of global solar radiation on horizontal surface, the monthly average temperature, the sun tracking settings to estimate the energy export to the grid, and the plant capacity factor [18]. Input data from the NASA global solar radiation database for the various locations are used in our study. For each location, a fixed slope at the latitude angle and facing south was considered. One side or two side tracking mode was not considered as it requires due maintenance following seasonal variations. The azimuth angle was taken as zero for all locations. As claimed in previous studies, the data from the satellite measurement of NASA may show some discrepancy with the ground measurement due to the level of dust in the air and other climatic conditions [19].

The selection of study locations is based on the access to the national grid and the availability of solar radiation data from NASA database. The analysis was completed for 35 locations and the final results are summarized in Table 1. The first column of the table includes the capacity factor of a 5 MW PV plant for each location. Capacity factor is the ratio of the actual output of a power plant over a period of time to its potential output if it had operated at the hypothetical maximum possible. The average value of PV

¹ Ethiopia has a unique calendar: 12 months, each 30 days; and the remaining 5 or 6 days make up the 13th month. As often the sun shines even in the rainy months, “13 months of sunshine” has remained an old motto of the nation.

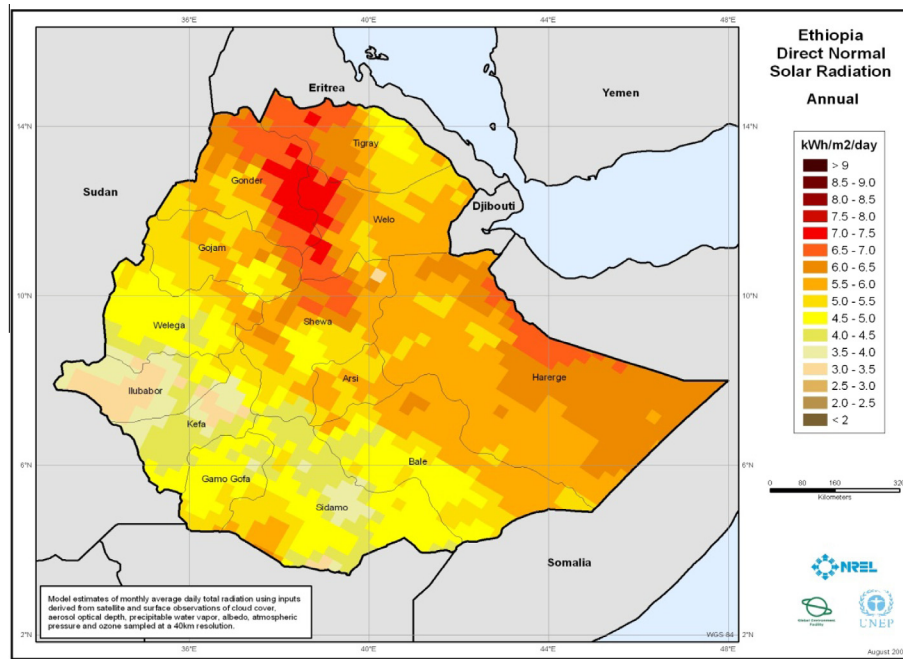


Fig. 1. Ethiopia's annual direct solar radiation potential (Source: [17]).

Table 1

Performance of 5 MW PV power plants in 35 locations of Ethiopia.

Place	Capacity factor (%)	Electricity to grid (MWh)	Annual GHG reduction (tCO ₂)	Litre of gasoline not consumed	Latitude	Longitude	Elevation (m)
Adigrat	20.1	8815.1	1254	538,809	14.3	39.5	1731
Arbaminch	19.3	8446.3	1202	516,466	6.0	37.6	1589
Asebe Teferi	20.6	9044.7	1287	552,988	9.1	40.9	1000
Asossa	18.9	8299.1	1181	507,443	10.1	34.5	877
Assela	19.8	8673.3	1234	530,215	8.0	39.1	2589
Awassa	20	8778.8	1249	536,661	7.1	38.5	1983
Bahir Dar	20.7	9062	1289	553,847	11.6	37.4	2070
Bati	20	8756.2	1246	535,372	11.2	40	720
Bedele	19.2	8388.5	1193	512,599	8.5	36.4	1808
Debre Birhan	20.2	8835.9	1257	540,098	9.7	39.5	2182
Debre Markos	20.6	9042.9	1286	552,558	10.3	37.7	2195
Debre Tabor	21.3	9331.2	1327	570,175	11.9	38.0	2219
Dessie	20.6	9025.6	1284	551,699	11.1	39.6	2286
Dilla	19.9	8736.3	1243	534,083	6.4	38.3	2109
Dire Dawa	20.3	8892.5	1265	543,535	9.6	41.9	1433
Dodolla	19.0	8301.6	1181	507,443	7.0	39.2	2135
Dubti	20.3	8878.8	1263	542,676	11.7	41.7	472
Fiche	21.0	9185	1307	561,582	9.8	38.7	2330
Gambella	18.1	8026.1	1142	490,686	8.3	34.6	952
Gewane	20.2	8860.9	1260	541,387	10.2	40.7	798
Gondar	21	9206.8	1310	562,871	12.6	37.5	1950
Harer	20.6	9003.1	1281	550,410	9.3	42.1	1563
Hosaena	19.5	8544.7	1216	522,481	7.6	37.9	1926
Jimma	18.1	7933.7	1129	485,100	7.7	36.8	1942
Jinka	19.2	8397.3	1195	513,458	5.7	36.7	942
Kemisse	21	9185.7	1307	561,582	10.7	39.9	2391
Mekelle	20.7	9070.9	1290	554,277	13.5	39.5	1911
Metu	18.6	8138.7	1158	497,560	8.3	35.6	1632
Nazret	20.7	9050.6	1287	552,988	8.6	39.3	1869
Negele	18.3	8017.2	1140	489,826	5.3	39.6	1402
Nekemte	20.2	8844.7	1258	540,528	9.1	36.5	1592
Sawla	19.1	8357.5	1189	510,880	6.3	36.9	1225
Shakiso	19.1	8378.9	1192	512,169	5.8	38.9	1647
Tepi	17.7	7730.8	1100	472,639	7.2	35.4	1719
Yabelo	19.2	8332.4	1198	514,747	4.9	38.1	1483

power plant capacity factor of the different locations is 19.8%. Among the locations considered Tepi has the minimum value of electricity feed into the grid (7730.8 MWh/year) while Debre

Tabor has the highest annual electricity production (9331.2 MWh/year). Since the northern part of the country (where Adigrat, Debre Tabor, Dessie, Kemisse, ... are located) is mountainous and

on high altitude, there is less precipitation and more solar radiation can reach the surface [16]. In this region the solar radiation power density is reported to exceed 230 W/m^2 [16]. Detail geographic differences and impact of weather conditions on the solar energy potential throughout the country is provided in [16, p. 52–57].

The base case electricity system for Ethiopia is taken from the current energy mix of hydropower (88.5%), gasoline (10.9%), and geothermal (0.6%). The GHG emission factor of the baseline electricity mix (fuel type) in Ethiopia is calculated to be $0.142 \text{ tCO}_2/\text{MWh}$. The average annual carbon reduction of the different locations considered in the study is 1234.3 tCO_2 , which is equivalent to 530,338 litres of gasoline not consumed (see further explanation under the section “GHG reduction”).

Overall, it can be inferred that Ethiopia has a high potential for both off-grid and on-grid PV system utilization [3]. The feasibility study of a 5 MW proposed on grid PV system on the outskirts of Addis Ababa is discussed in the next section.

Feasibility study of grid-connected 5 MW PV plant in Addis Ababa

There has been a recently growing interest from the Government of Ethiopia (GoE) in exploring the potential of PV grid power plants. Among the on-going efforts, a consulting firm from China was recruited to study the solar and wind resource potential of the country for installing such plants. The consulting team recommended four PV power station sites as part of its pre-feasibility study report [16]. However, the report did not provide any detail economic feasibility study. The consultants merely suggested that a 5 MW PV system would cost about USD 13 million [16, p. 160]. Looking at the trend, the GoE may embark on one of these projects. Recently, the GoE has announced a plan to establish 300 MW solar parks in the country [20].

Here we focus on a typical demonstration PV plant of 5 MW size to be installed around the Entoto Mountain, located on the outskirts of Addis Ababa (see Fig. 2). This site is also recommended by the aforementioned consulting team.

Besides interviewing the relevant officers in Ethiopia, we gathered data from the National Meteorological Agency of Ethiopia; but what was available and delivered to the author was only the average daily sunshine hours data. Based on the data gathered from the agency, the last 10 years average daily sunshine hours in Addis Ababa is depicted in Fig. 3. The average sunny hours per day is about 6.6 h.

Providing geographic location data (latitude) to HOMER software provides radiation values from clearness indices, and vice versa. The clearness index is a measure of the clearness of the

atmosphere. It is the fraction of the solar radiation that is transmitted through the atmosphere to strike the surface of the Earth. HOMER displays the monthly average radiation and clearness index of the baseline data and can create synthesized values using the Graham algorithm. The baseline data is a one-year time series representing the average global solar radiation on the horizontal surface. Detail description of this software is available in [21].

Using HOMER software, the daily radiation and clearness index of the proposed site is presented in Fig. 4. The annual average daily radiation is 5.07 kWh/m^2 (tilted equal to the latitude), and the average clearness index is 0.513. The Data map (DMap) generated by HOMER also shows the daily and seasonal patterns (Fig. 5). As per the ground measurement data documented in the RETScreen Climate Database, the average daily solar radiation (horizontal)

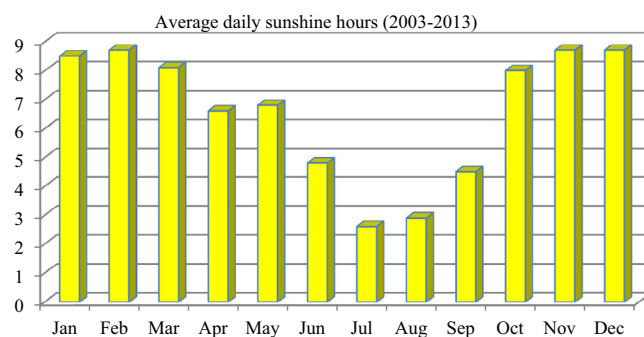


Fig. 3. Average sunshine hours in Addis Ababa (Source: National Meteorology Agency of Ethiopia).

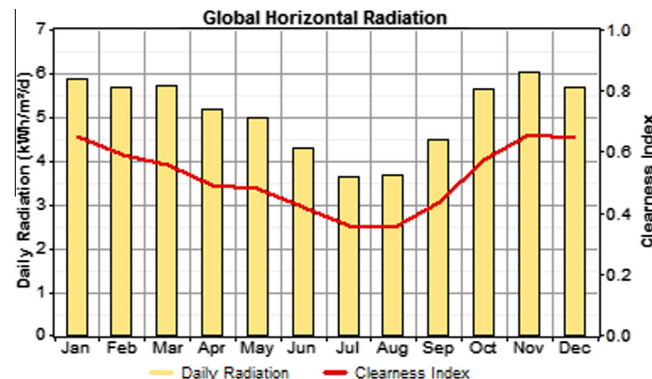


Fig. 4. Daily radiation and clearness index for the proposed site.

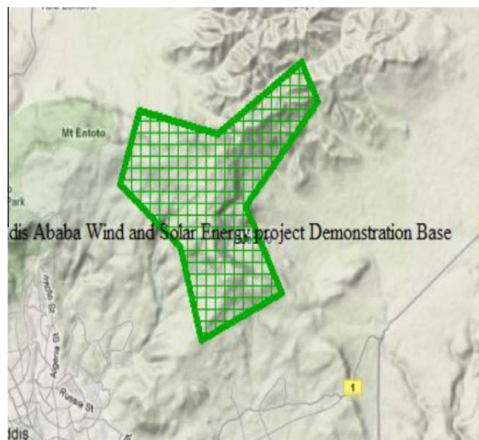


Fig. 2. Proposed site for Addis Ababa PV power plant [16, p. 154–155].

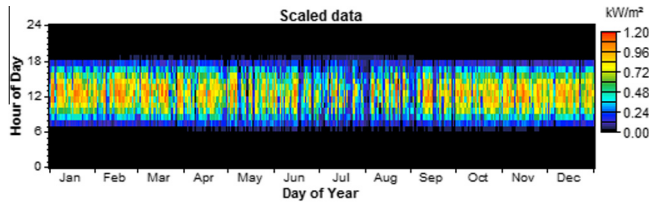


Fig. 5. Monthly average solar radiation.

for Addis Ababa is 4.99 kWh/m^2 . The RETScreen uses either the NASA global satellite data locations or ground-based meteorological data stations. The daily radiation values presented here (Fig. 4 and Table 2) showed some differences to that of the NASA satellite measurement. For our analysis, we used the ground-based data (Table 2). In Ethiopia, June to September is the rainy season and as a result, the monthly clearness index varies between 0.660 in November (dry season) and 0.353 in August (rainy season).

Proposed PV system

For the proposed grid-tied 5 MW solar PV system, we selected Mono-Si-HIP-215NKHA5 PV module which is manufactured by Sanyo. This PV module was selected for the study as it is one of the efficient available PV modules in the market. A detail specification of the PV module is given in Table 3. All figures are stated under the standard reference condition, i.e. total irradiance of 1000 W/m^2 ; air mass of 1.5; and cell temperature of 25°C . The total number of PV modules required for the power plant is estimated by dividing the required total watt peak rating to the module peak rated output [21] as shown in Eq. (1). Detail method of designing solar PV plant is available in [22].

Number of PV modules required

$$= \frac{\text{Total watt peak rating}}{\text{PV module peak rated output}} \quad (1)$$

The total number of PV modules required for the proposed system would be 23,256 panels. The area required for the installation of the PV modules would amount to $34,364 \text{ m}^2$ (this can be estimated by multiplying the number of required modules by the unit area required for a module). The panels are to be installed in fixed structure and inclined at an angle equal to the site latitude while the azimuth angle is taken as zero. In selecting an inverter for the system, we reviewed previous studies and current market trend. A previous feasibility of a 5 MW PV plant in Saudi Arabia considered an inverter with a capacity of 4,750 kW and 95% efficiency [23]. Another study in Egypt suggested two identical

Table 2

Daily solar radiation in Addis Ababa (Source: [18]).

Month	Daily solar radiation – horizontal $\text{kWh/m}^2/\text{d}$	Daily solar radiation – tilted (at the latitude angle) $\text{kWh/m}^2/\text{d}$
January	5.48	5.87
February	5.47	5.70
March	5.64	5.71
April	5.27	5.19
May	5.17	4.99
June	4.47	4.29
July	3.77	3.65
August	3.73	3.66
September	4.50	4.50
October	5.47	5.64
November	5.65	6.02
December	5.27	5.68
Annual	4.99	5.07

Table 3

Specification of PV module (Source: [26]).

Item	Specification
Manufacturer	Sanyo
PV Model	HIT Power 215 N or HIP-215NKHA5
Module Efficiency	17.1%
Maximum Power Voltage (V _{mp})	42.0 V
Maximum Power Current (I _{mp})	5.13 A
Open Circuit Voltage (V _{oc})	51.6 V
Short Circuit Current (I _{sc})	5.61 A
Temperature Coefficient (P _{max})	−0.336%/°C
Temperature Coefficient (V _{oc})	−0.143 V/°C
Temperature Coefficient (I _{sc})	1.96 mA/°C
Cell Efficiency	19.3%
Maximum System Voltage	600 V
Warranted Tolerance (−/+)	−0%/+10%
Module Area	1.26 m ²
Dimensions L × W × H	1580 mm × 798 mm × 46 mm
Weight	16 kg
Ambient Operating Temperature	−4 °F to 115 °F (−20 °C to 46 °C)

Table 4

Specification of ABBPVS800 (1 MW) inverter (Source: [25]).

Parameter		Specification
DC input	DC voltage range, mpp (UDC, mpp)	600–850 V
	Maximum DC voltage (Umax(DC))	1100 V
	Maximum DC current (Imax(DC))	1710 A
AC output	Nominal output voltage (UN(AC))	400 V
	Nominal AC current (IN(AC))	1445 A
	Output frequency	50/60 Hz
	Harmonic distortion, current	<3%
Other	Peak efficiency	98.8%

inverters with rated capacity of 4,750 kW and 95% efficiency for a 10 MW grid-tied PV plant [24]. In this study, we considered five identical ABB PVS800 central inverters of size 1000 kW each and maximum efficiency of 98.8% (see Table 4). However, in the analysis, we considered 96% efficiency for the sake of more practicality. The ABB inverters are selected considering their market availability, cost-effectiveness and high efficiency.

Financial analysis

Studying the economic viability of PV power plant benefits and informs both investors and policy makers. Here we conducted the financial feasibility study of the proposed 5 MW PV power plant using RETScreen version 4.0 software. As part of the RETScreen Clean Energy Project Analysis Software, a Financial Analysis worksheet is provided for each project evaluated. The financial analysis worksheet contains six sections: Financial parameters, annual income, project costs and savings/income summary, financial viability, yearly cash flows and cumulative cash flows graph [18].

For our analysis, the initial investment for the project, the annual operation and maintenance costs (periodic costs), the useful life period of the PV panel (25 years), and the electricity output sale to the grid (to EEPCo) were considered. The initial costs for the analysis included feasibility study costs, development and engineering costs, power system and balance of system (BOS) costs. Besides the initial costs, annual and other periodic costs were included (Table 5).

With the current PV modules market in Ethiopia, the price of a solar panel of Mono-Si-HIP-215NKHA5 is estimated to be 850 USD. The total cost of PV modules would then be 19,767,600 USD (385,468,200 ETB²). As such large sized PV grid systems are not yet introduced in Ethiopia and nearby countries, gathering cost data

² 1 USD ≈ 19.5 ETB (Ethiopian Birr).

Table 5
Initial costs and other parameters.

	Unit	Value
<i>Initial costs</i>		
Feasibility study, development and engineering cost	Percent of total initial cost	0.6% [24,27]
Power system cost	USD/Wp	3.954 USD/Wdc (about 72% of initial cost)
BOS costs	Percent of total initial cost	24.4% (inverter cost, about 22% of initial cost)
Miscellaneous	Percent of total initial cost	3%
<i>Periodic costs</i>		
O and M cost	Annual cost	0.02 USD/Wdc [28]
Inverter replacement cost	Periodic (every five year)	\$102,565 [24]
<i>Other assumptions</i>		
Inflation rate	Percent	8.1% [29]
Discount rate	Percent	10%
Debt interest rate	Percent	5.0 [30]
Debt period	Period	15 years
Energy cost increase rate	Percent	5
Analysis period	No. of years	25

remains challenging. A previous study of 10 MW grid-tied PV system in Egypt, for instance, estimated that 24% of the initial cost goes to the BOS cost, while 70% is the PV cost, and other miscellaneous costs including feasibility study cost sum up to 5% [24]. Considering the market trend and estimated cost of system components, we assumed 24.4% cost is that of the BOS cost, and the main cost category is that of the module cost. Further cost input data and estimates collected through expert interviews and document reviews were used in the analysis (see Table 5).

The RETScreen presents the main financial indicators and analysis results including the net present value (NPV), internal rate of return (IRR), benefit-cost ratio (B–C), payback period and production cost. NPV is used to determine the present value of an investment. Employing NPV method, the present value of all cash inflows is compared against the present value of all cash outflows associated with the project. The difference between the present values of these cash flows, called the NPV, determines whether or not the project is generally a financially acceptable investment. Positive NPV is an indicator of a potentially feasible project (see Eq. (2) for the basic formula used in NPV calculation). The RET Screen model calculates the NPV using the cumulative after-tax cash flows.

$$NPV = -C_0 + \sum_{i=1}^T \frac{C_i}{(1+r)^i} \quad (2)$$

C_0 – initial investment, C – cash flow, r – discount rate, T – time.

The other financial feasibility indicator is the internal rate of return (IRR) which represents the true interest yield provided by the project equity over its life. IRR is the return on investment or the time-adjusted rate of return and is calculated by finding the discount rate that causes the NPV of the project to be equal to zero. If the IRR is equal to or greater than the required rate of return, then the development of the project will likely be considered financially acceptable. The model employed by RETScreen also calculates the net Benefit–Cost (B–C) ratio, which is the ratio of the net benefits to costs of the project. Net benefits represent the present value of annual income and savings less annual costs, while the cost is defined as the project equity. B–C ratios greater than 1 are indicative of profitable projects.

In this feasibility analysis of the proposed 5 MW grid-connected PV system, considering the on-going investment practice in Ethiopia, a 50% investment loan with a 5% debt interest rate was

assumed. With these input parameters, it was found that the project will have a NPV of 1,510,636 USD (29,457,403 ETB) and pre-tax IRR of 8.8%. As Ethiopia has not yet approved a feed-in tariff (FIT) law for solar power generation, we assumed a rate of 200 USD/MWh (taken from the FIT of neighboring Kenya [31]). According to the analysis, the equity payback period is 14.5 years and its B–C ratio is 1.12. With all these financial indicators, it may be concluded that the investment is economically viable but not sufficiently attractive for commercial investors.

GHG reduction

The RETScreen calculates the annual net GHG reduction in equivalent tonnes of CO₂ per year (tCO₂/yr) resulting from the implementation of the proposed case system instead of the base case, or baseline system. It is determined by the formula shown in Eq. (3). The RETScreen also compares the net annual GHG emission reduction with units that are easier to conceptualize (e.g. Cars & light trucks not used, litres of gasoline not consumed, etc). These numbers are based on North American energy use patterns and are published by the Environmental Protection Agency (EPA), 2009 [18]. The model is an easier way of estimating the GHG emission reduction of the proposed project but requires the user to be vigilant enough in defining the base case scenario [18].

Annual GHG emission reduction (tCO₂)

$$= (\text{Base case GHG emission factor (tCO}_2\text{/MWh)} \\ - \text{Proposed case GHG emission factor (tCO}_2\text{/MWh)}) \\ \times \text{End use energy delivered (MWh)} \quad (3)$$

The proposed 5 MWp PV system in this study can generate up to 7658 MWh electricity annually that can be fed into the grid. The GHG emission factor of the baseline electricity mix (fuel type) in Ethiopia is 0.142 tCO₂/MWh. The net annual GHG emission reduction of the proposed system would be 1089 tCO₂, which is equivalent to 467,913 litres of gasoline not consumed.

Sensitivity analysis

One of the assumed input parameter for the feasibility study is the FIT. FIT is a critical factor particularly for attracting investors to the emerging solar power market. As depicted in Table 6, the sensitivity analysis result on NPV analysis shows significant variations due to the variations of such input data. If FIT is reduced by 15%, NPV would change from 1,510,636 USD (29,457,403 ETB) to –2,554,312 USD (–49,809,075 ETB). Variations on the debt interest rate, discount rate and inflation rate also significantly affect the NPV. For instance, a 10% debt interest rate changes the energy production cost from 188.85 USD/MWh (3,682.6 ETB/MWh) to 216.82 USD/MWh (4,227.95 ETB/MWh). A 15% reduction in debt interest rate can also change the NPV by about 520,000 USD (10,140,000 ETB) (Table 7). Further sensitivity analyses show that FIT takes the highest influence on the NPV. For example, an increase of FIT by 15% results in about 270% increment of NPV which is not attainable by any increment or reduction of other input parameters by the same percentage. Introducing attractive FIT law (at least comparable to Kenyan FIT) for solar electricity in Ethiopia could be a starting point to pave the way for private sector investment in the sector.

Comparing to other cases

The energy production cost for the proposed PV system was calculated to be 0.197 USD/kWh. A study on the feasibility of a 10 MWp solar PV plant in Kenya showed that the levelized cost

Table 6

Sensitivity analysis results of NPV for the proposed system (with variation of FIT).

FIT USD/MWh Change	%	Initial costs (USD)				
		17,642,575.85 –30%	21,423,127.79 –15%	25,203,679.74 0%	28,984,231.69 15%	32,764,783.69 30%
140.00	–30	278,886.27	–3,170,186.44	–6,619,259.15	–10,068,331.86	–13,517,404.56
170.00	–15	4,343,833.88	894,761.17	–2,554,311.54	–6,003,384.25	–9,452,456.95
200.00	0	8,408,781.48	4,959,708.78	1,510,636.07	–1,938,436.64	–5,387,509.35
230.00	15	12,473,729.09	9,024,656.39	5,575,583.68	2,126,510.97	–1,322,561.74
260.00	30	16,538,676.70	13,089,603.99	9,640,531.29	6,191,458.58	2,742,385.87

Table 7

Sensitivity analysis results of NPV for the proposed system (with variation of debt interest rate).

Debt interest rate (%)	%	Initial costs (USD)				
		17,642,576 –30%	21,423,128 –15%	25,203,680 0%	28,984,232 15%	32,764,784 30%
3.50	–30	9,127,385	5,832,299	2,537,213	–757,874	–4,052,960
4.25	–15	8,772,859	5,401,803	2,030,747	–1,340,309	–4,711,365
5.00	0	8,408,781	4,959,709	1,510,636	–1,938,437	–5,387,509
5.75	15	8,035,389	4,506,304	977,218.7	–2,551,867	–6,080,952
6.50	30	7,652,933	4,041,892	430,852.2	–3,180,188	–6,791,228

of electricity (LCOE) from PV is estimated to be about 0.21 USD/kWh [15]. Another recent study in Nigeria analyzed the technical and economic performance of an 80 kW solar PV grid connected system (contributing 40.4%) in combination with a 100 kW power from the grid and showed that the LCOE was about \$0.103/kWh [32]. Looking at such cases, the proposed system cost in Ethiopia falls within the range of LCOE in the region. But our previous study identified that the policy makers in Ethiopia believe that solar is too costly and not a viable option [3]. The current electricity tariff in Ethiopia is highly subsidized and one of the lowest in Africa. The tariff depends on the monthly energy consumption and varies among user classification. For instance, according to the 2013 tariff of EEPCo, customers with a monthly consumption up to 50 kWh pay a flat rate of 0.273 ETB/month (0.014 USD/month) plus a service charge of either 1.4 ETB/month (0.072 USD/month) for 0 to 25 kWh or 3.4 ETB/month (0.174 USD/month) for 26 to 50 kWh [33,34].

In general, as this paper aims to serve as an early contribution to the viability study of grid-tied PV systems in Ethiopia and neighboring countries, emphasis was placed on the variation of FIT and its impact on the financial viability of the PV system (Tables 6 and 7). With the global trend of PV technology cost reduction and the recent effort in assembly of solar panels locally in Ethiopia, a scenario of reduced initial costs may prevail. Such a scenario combined with favorable FIT legislation may make the PV on-grid investment more attractive. As suggested by a previous study [15], the FIT needs to serve as an instrument for attracting private investment in Ethiopia and other sub-Saharan African countries.

Concluding remarks

In this study, the grid-connected solar PV power generation potential of 35 locations in Ethiopia was examined. It was found in the study that the mean value that can be generated from a 5 MW PV plant in those locations is 8674 MWh/yr. The average value of PV power plant capacity factor of the different locations was also found to be 19.8%. A detail feasibility study for a grid-tied 5 MW PV system in Addis Ababa area was also conducted. It was found that 7658 MW can be generated annually from the system and the financial indicators showed that the investment is economically viable but not sufficiently attractive for commercial

investors. Moreover, it was found that a pre-feasibility study for such a proposed PV system underestimated the investment cost by almost 50%. As it requires high initial investment for joining the solar PV market, there is a need to provide incentives and create conducive investment environment. For a developing country like Ethiopia, too much reliance on hydropower is quite a risky strategy while possessing an array of other renewable energy resources. Besides the on-going huge state investment, some of the policy instruments recommended to attract private investors would be provision of tax incentives, financial loans, grants and formulation of favorable FIT laws. Finally, further economic viability study on PV grid-connected power plants across the potential sites, mainly when FIT law and legislation unfolds in the country, is recommended. Studies like ours suffer from the usual limitation of having assumed input parameters, and thus appraisal of upcoming grid-connected solar PV system projects is strongly recommended.

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